



Date: 12 / 07 / 26

Lab Practical #04:

Installation of Network Simulator (Packet Tracer) and Implement different LAN topologies.

Practical Assignment #04:

1. Create a simple network with switch and two or more pc. Also check connectivity between them using ping command or PDU utility.

- ✓ To set up a simple network and test connectivity, you can use a network simulation tool like Cisco Packet Tracer.

I. Building the Network Topology

- You will need three devices to create this network:
- **1 Switch:** Use a standard switch (e.g., Cisco 2960).
- **3 PCs:** Add two desktop computers (PC-0, PC-1 and PC-2).
- **Cables:** Use Straight-Through cables to connect the PCs to the switch.

Connections:

1. Connect **PC-0** (FastEthernet0 port) to **Switch** (FastEthernet0/1 port).
2. Connect **PC-1** (FastEthernet0 port) to **Switch** (FastEthernet0/2 port).
3. Connect **PC-2** (FastEthernet0 port) to **Switch** (FastEthernet0/3 port).

II. Configuring IP Addresses

- Devices need IP addresses on the same network subnet to talk to each other.
- Click on each PC, go to the Desktop tab, open IP Configuration, and assign the following details:
- PC-0 Setup:
 - ✓ IP Address: 192.168.1.1
 - ✓ Subnet Mask: 255.255.255.0
- PC-1 Setup:
 - ✓ IP Address: 192.168.1.2
 - ✓ Subnet Mask: 255.255.255.0
- PC-2 Setup:
 - ✓ IP Address: 192.168.1.3
 - ✓ Subnet Mask: 255.255.255.0

Date: 12 / 07 / 26**III. Testing Connectivity**

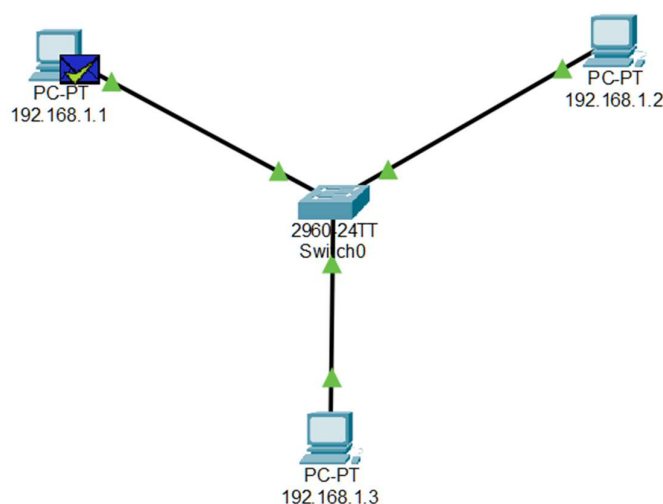
- Once the link lights on the switch turn from amber to green, you can test the connection using two different methods

Method A: Using the Ping Command (CLI)

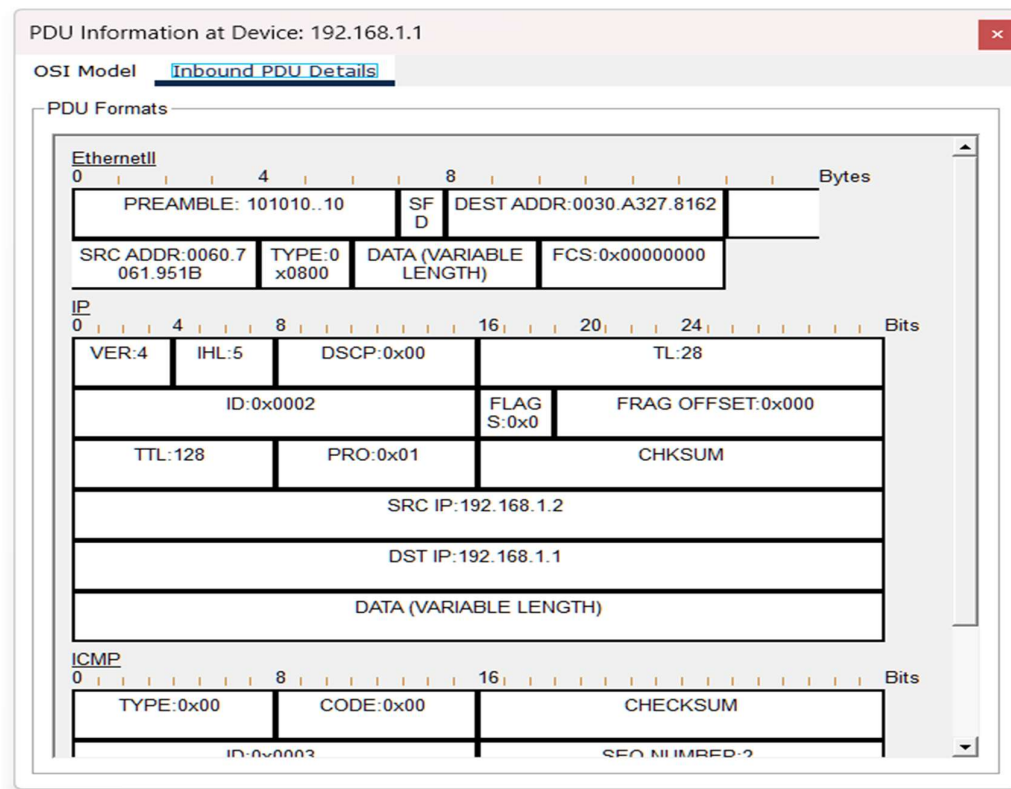
- Click on PC-0 and open the Command Prompt from the Desktop tab.
- Type the following command to test the connection to PC-1 and press Enter:
ping 192.168.1.2
- Successful Result: You should see four replies indicating a successful connection
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128 ...
Ping statistics: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Method B: Using the PDU Utility (Visual GUI)

- Locate the Add Simple PDU tool (it looks like a closed envelope icon) on the right-hand or top toolbar.
- Click the envelope tool, then click on **PC-0** (the source).
- Move your mouse and click on **PC-1** (the destination).
- Look at the Simulation/Realtime panel in the bottom-right corner. The status tracking window will change to say Successful, proving that the data packet successfully traveled through the switch and back.



Date: 12 / 07 / 26



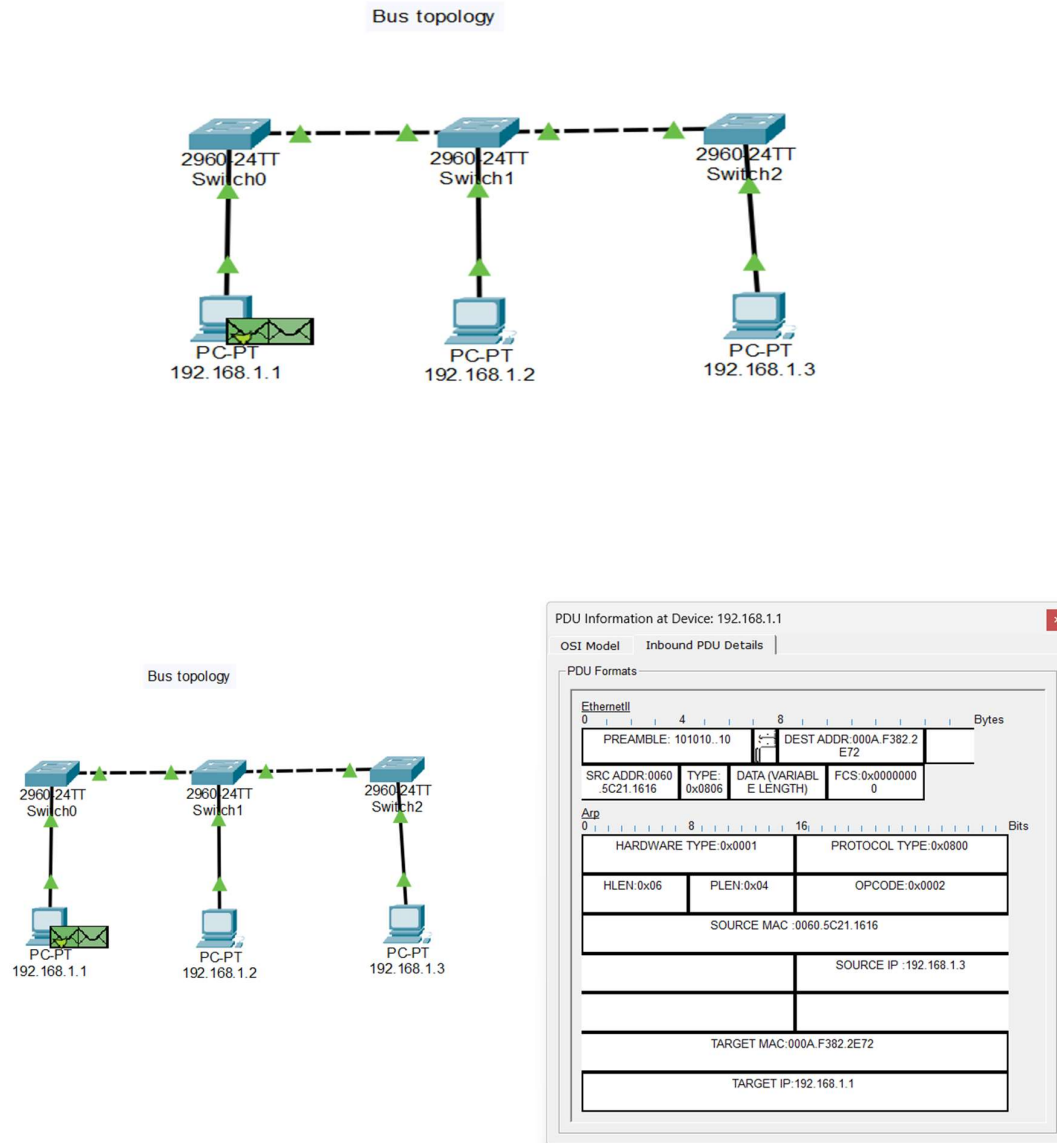
2. Implement different topologies in packet tracer.

a. Bus

- A linear network structure where all devices share a single backbone cable. In Packet Tracer, this is simulated by daisy-chaining switches.
- **Design Setup:** Add 3 Switches (2960) side-by-side. Connect Switch0 to Switch1 and Switch1 to Switch2 using Copper Cross-Over cables. Connect one PC to each switch using Copper Straight-Through cables.

Device Name	Interface	IP Address	Subnet Mask
192.168.1.1	FastEthernet0	192.168.1.1	255.255.255.0
192.168.1.2	FastEthernet0	192.168.1.2	255.255.255.0
192.168.1.3	FastEthernet0	192.168.1.3	255.255.255.0
Switch0	FastEthernet()	192.168.1.1	255.255.255.0
Switch1	FastEthernet()	192.168.1.2	255.255.255.0
Switch2	FastEthernet()	192.168.1.3	255.255.255.0

Date: 12 / 07 / 26



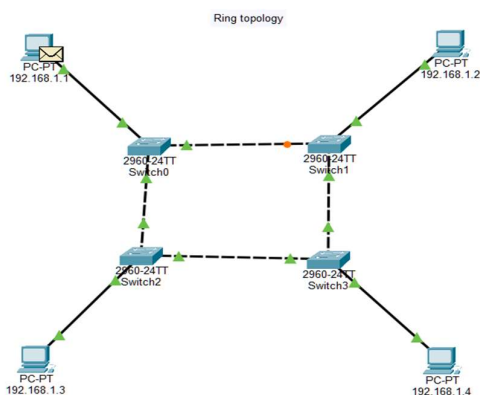
b. Ring

- A circular network architecture where data travels sequentially from one node to the next until it reaches its destination.
- **Design Setup:** Add 3 Switches (2960) arranged in a triangular/circular shape. Connect Switch0 to Switch1, Switch1 to Switch2, and Switch2 back to Switch0 using Copper Cross-Over cables. Connect one PC to each individual switch. (*Note: Spanning Tree Protocol will block one link by default to prevent loops, which is normal behavior in Packet Tracer*).

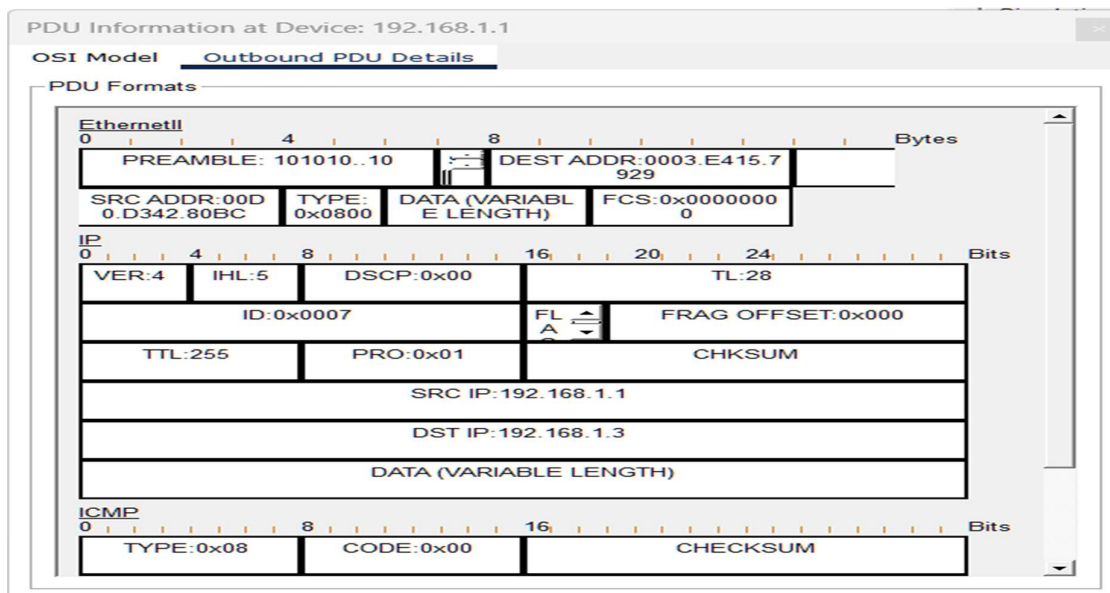


Date: 12 / 07 / 26

Device Name	Interface	IP Address	Subnet Mask
192.168.1.1	FastEthernet0	192.168.1.1	255.255.255.0
192.168.1.2	FastEthernet0	192.168.1.2	255.255.255.0
192.168.1.3	FastEthernet0	192.168.1.3	255.255.255.0
192.168.1.4	FastEthernet0	192.168.1.4	255.255.255.0
Switch0	FastEthernet0	192.168.1.1	255.255.255.0
Switch1	FastEthernet0	192.168.1.2	255.255.255.0
Switch2	FastEthernet0	192.168.1.3	255.255.255.0
Switch3	FastEthernet0	192.168.1.4	255.255.255.0



Date: 12 / 07 / 26

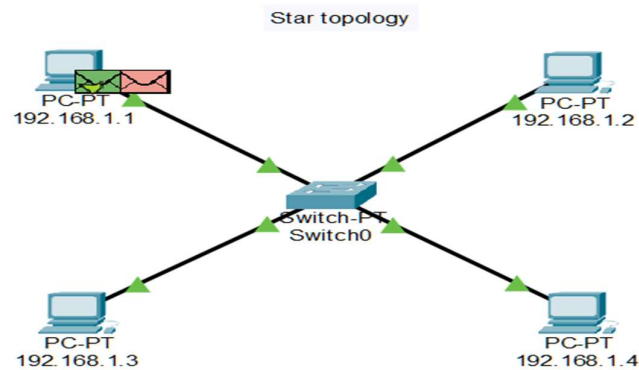


c. Star

- The most common modern deployment where all peripheral nodes connect directly to a single centralized network hub or switch.
- **Design Setup:** Add 1 Central Switch (2960). Add 3 or 4 PCs around it. Connect each PC directly to the central switch using Copper Straight-Through cables.

Device Name	Interface	IP Address	Subnet Mask
192.168.1.1	FastEthernet0	192.168.1.1	255.255.255.0
192.168.1.2	FastEthernet0	192.168.1.2	255.255.255.0
192.168.1.3	FastEthernet0	192.168.1.3	255.255.255.0
192.168.1.4	FastEthernet()	192.168.1.4	255.255.255.0
Switch0	FastEthernet()	192.168.1.1	255.255.255.0

Date: 12 / 07 / 26



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

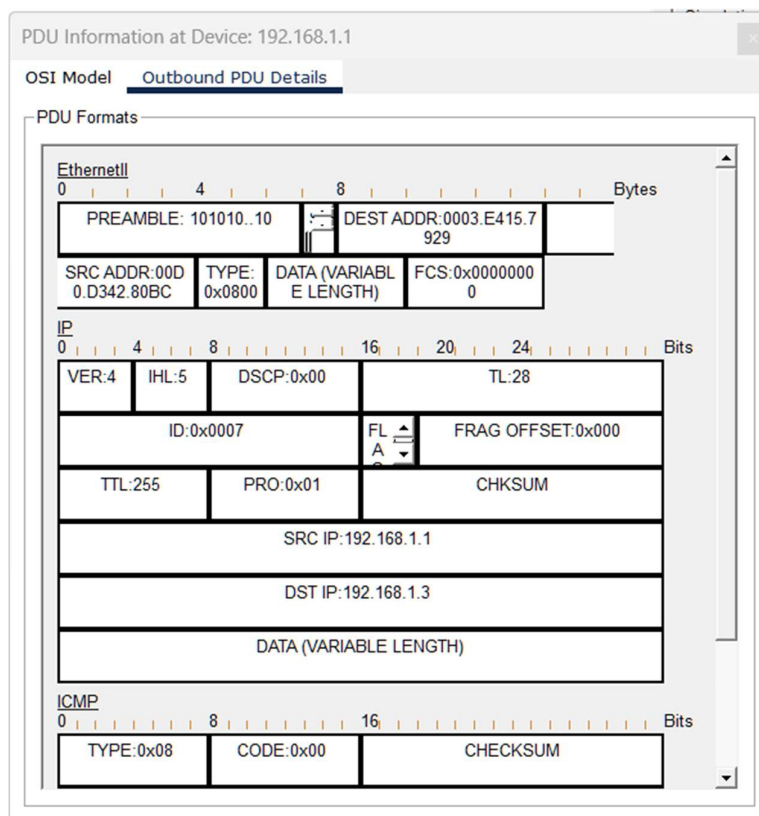
Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Date: 12 / 07 / 26

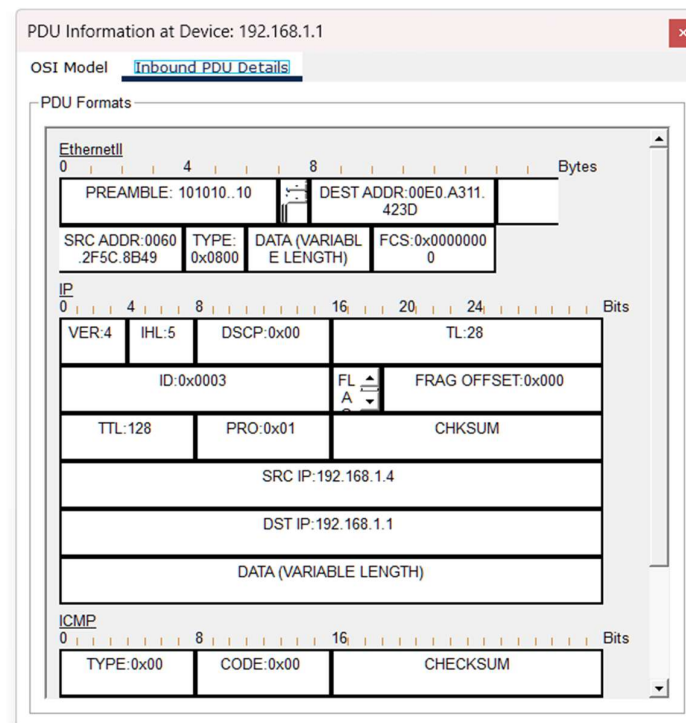
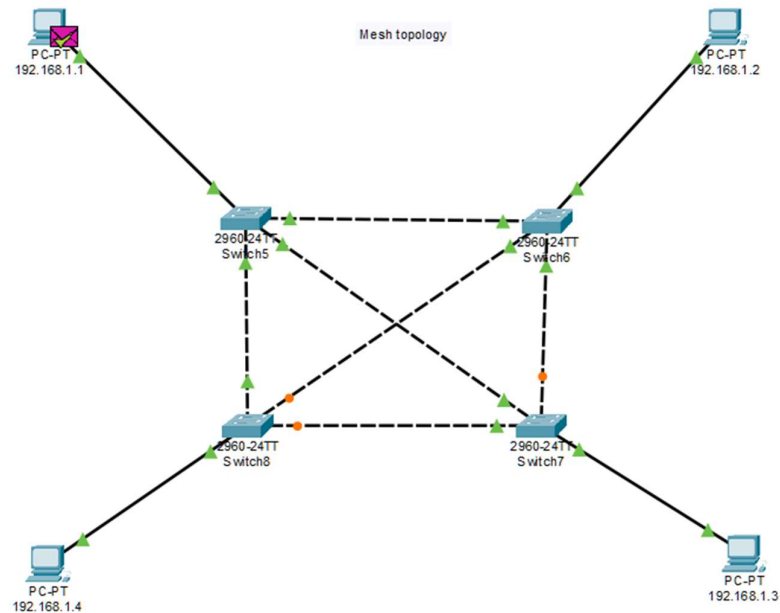


d. Mesh

- A highly redundant system where every single node is connected directly to every other node in the network framework.
- **Design Setup:** Add 3 Switches (2960). Connect them so that every switch has a point-to-point path to every other switch (Switch0-Switch1, Switch1-Switch2, Switch0-Switch2). Connect a PC to each switch.

Device Name	Interface	IP Address	Subnet Mask
192.168.1.1	FastEthernet0	192.168.1.1	255.255.255.0
192.168.1.2	FastEthernet0	192.168.1.2	255.255.255.0
192.168.1.3	FastEthernet0	192.168.1.3	255.255.255.0
192.168.1.4	FastEthernet0	192.168.1.4	255.255.255.0
Switch5	FastEthernet0	192.168.1.6	255.255.255.0
Switch6	FastEthernet0	192.168.1.7	255.255.255.0
Switch7	FastEthernet0	192.168.1.8	255.255.255.0
Switch8	FastEthernet0	192.168.1.9	255.255.255.0

Date: 12 / 07 / 26

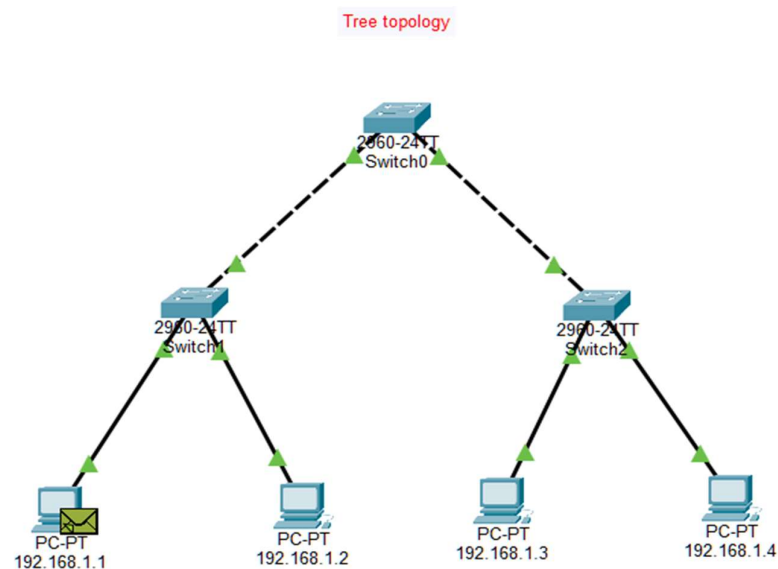


Date: 12 / 07 / 26

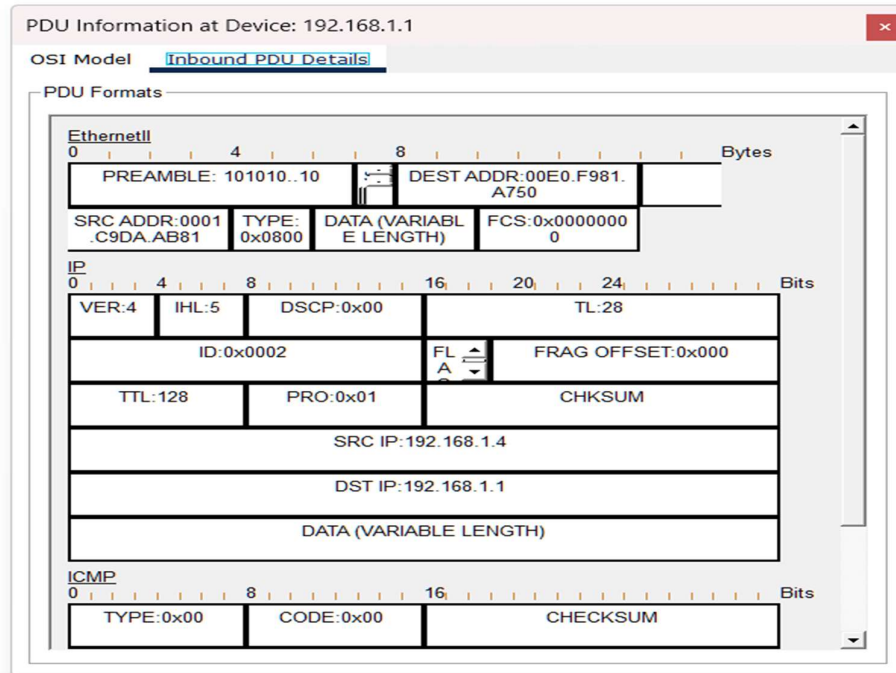
e. Tree

- A hierarchical topology that integrates multiple star topologies onto a single bus backbone hierarchy.
- **Design Setup:** Add 1 Root Switch at the top layer. Connect it downwards to 2 Secondary Switches (Distribution layer) using cross-over cables. Connect 2 PCs to each secondary switch using straight-through cables.

Device Name	Interface	IP Address	Subnet Mask
192.168.1.1	FastEthernet0	192.168.1.1	255.255.255.0
192.168.1.2	FastEthernet0	192.168.1.2	255.255.255.0
192.168.1.3	FastEthernet0	192.168.1.3	255.255.255.0
192.168.1.4	FastEthernet0	192.168.1.4	255.255.255.0
Switch0	FastEthernet0	192.168.1.1	255.255.255.0
Switch1	FastEthernet0	192.168.1.2	255.255.255.0
Switch2	FastEthernet0	192.168.1.3	255.255.255.0



Date: 12 / 07 / 26





Date: 12 / 07 / 26

3. Switch Configuration via CLI

- To configure a management IP address on any of your topology switches or change a device hostname, access the switch CLI tab and input the following configuration commands sequence:

Switch> enable

Switch# configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)# hostname Central-SW

Central-SW(config)# interface vlan 1

Central-SW(config-if)# ip address 192.168.1.1 255.255.255.0

Central-SW(config-if)# no shutdown

Central-SW(config-if)# exit

Central-SW(config)# exit

Central-SW# copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]